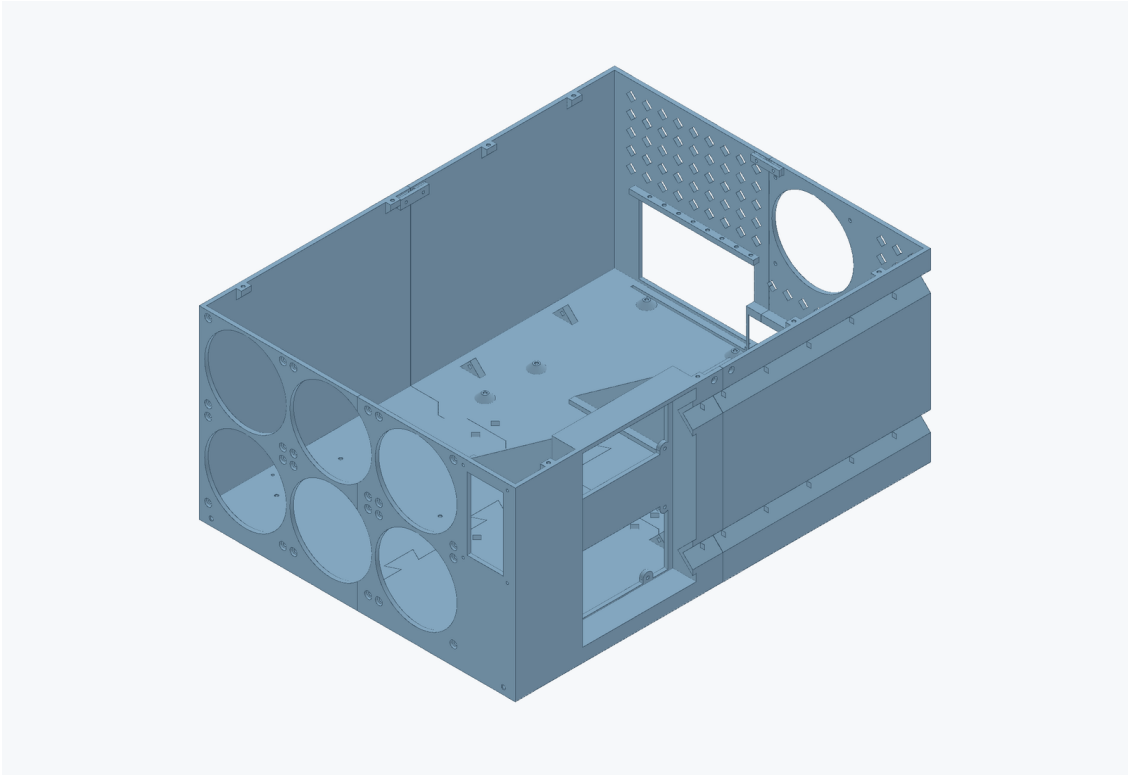
CHASSIS ASSEMBLY

Four sections. One chassis.

Illustrated bench instructions for the printed shell and its mounting features.



Actual Step16 CAD. The chassis is shown without duct parts, electronics, rods or a lid. All illustrations use the existing model; no production geometry was changed.

Finished envelope	Printed structure	Your rod arrangement
17.5 x 23 x 10.25 in 444.5 x 584.2 x 260.35 mm Width x depth x height	4 main sections 4 underside bridges 3 upper straps	2 crosswise rods, front/rear 2 longitudinal rods Middle crosswise omitted

As-built notes retained

Your rods are longer than the original 406.4 mm specification. The middle crosswise rod is omitted so your two longitudinal rods can fit.

The PCIe bracket top flange lacks rearward clearance in the printed chassis. That interference is recorded in the card-fit instructions; this guide does not prescribe a geometry change.

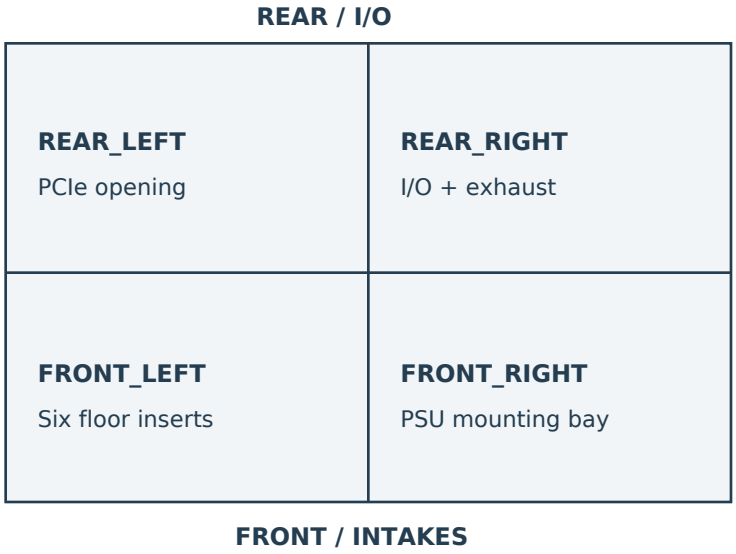
Build order: identify parts and hardware; install heatsets; join the quadrants; fit bridges, straps and rods; install fans; fit motherboard/cards and PSUs; route cables and inspect.

Work with the base supported on a flat bench. Finish all underside fasteners before electronics make them difficult to reach. This custom 6U project has no modeled rack ears; use a shelf supporting its base.

PARTS AND ORIENTATION

Lay out the eleven prints

Left and right are always named while standing at the front and looking toward the rear.



MAIN SECTIONS

One of each quadrant

CONNECTORS

4 base bridges

3 upper straps

Main filenames: 6U_Step16_FRONT_LEFT / FRONT_RIGHT / REAR_LEFT / REAR_RIGHT_Print.stl. The other three quadrants and all seven connectors remain compatible with Step15; keep already printed compatible pieces.

Connecting part names	Qty	Position
BRIDGE_X_FRONT / BRIDGE_X_REAR	2	Across the left/right seam in the floor.
BRIDGE_Y_LEFT / BRIDGE_Y_RIGHT	2	Across the front/rear seam in the floor.
STRAP_LEFT / STRAP_RIGHT	2	High on the sidewalls at the front/rear seam.
STRAP_REAR	1	Across the rear-wall centre seam.

Structural hardware - sort by length

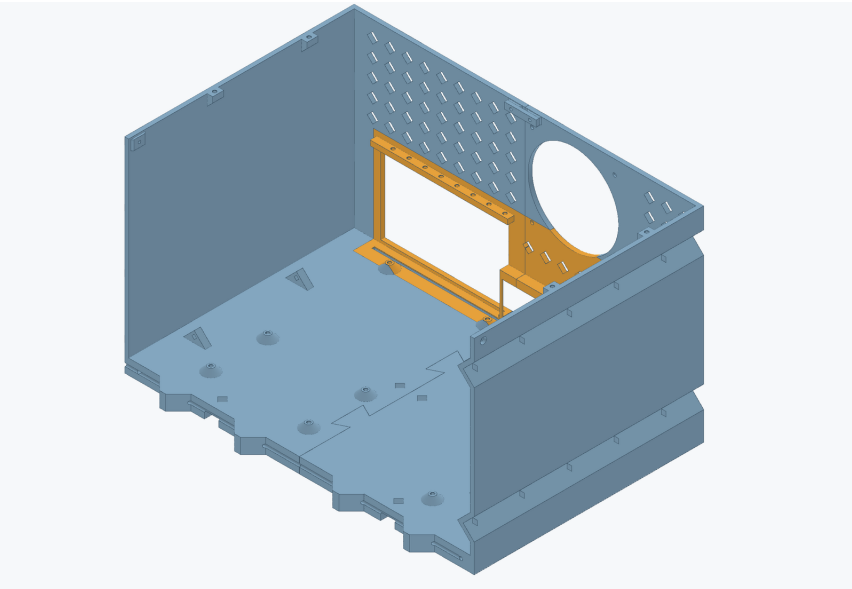
Connection	M3 screw	Nuts	Washers
Four base bridges	8 x 14 mm	8	16
Three upper straps	6 x 12 mm	6	12
Two longitudinal rod stops	2 x 18 mm	2	4

Use 0.5 mm washers, OD no more than 7 mm, and M3 nuts about 5.5 mm across flats x 2.4 mm thick. Structural socket heads must fit the printed recesses (model allowance: no more than 6 mm diameter x 3 mm high). A washer goes under each screw head and each nut.

PREPARATION

Install the chassis heatsets

Do this while the parts are easy to support and before electronics are installed.



Rear chassis CAD. Gold highlights the I/O and card-mounting region. The motherboard standoffs are already part of the floor; no separate standoff prints are required.

Location	Count	Purpose
Motherboard standoff tops	9	M3 motherboard screws.
Rear PCIe retention ledge	8	Card bracket retention positions.
Top rim / lid bosses	8	Future removable lid.
Front-left floor	6	Reserved support mounting points; no duct assembly in this guide.

31 pockets total. They use ruthex RX-M3Sx4.0: M3 thread, 4 mm insert length. Modeled pockets are 4 mm diameter x 5.2 mm deep with a small lead-in. Already fitted inserts count toward the total; unused lid/floor inserts may be deferred.

1. Clean first. Remove strings, loose debris and obvious first-layer lips from mating faces. Inspect the dovetails, locating tongues, rod entrances and recessed nut pockets without enlarging them indiscriminately.

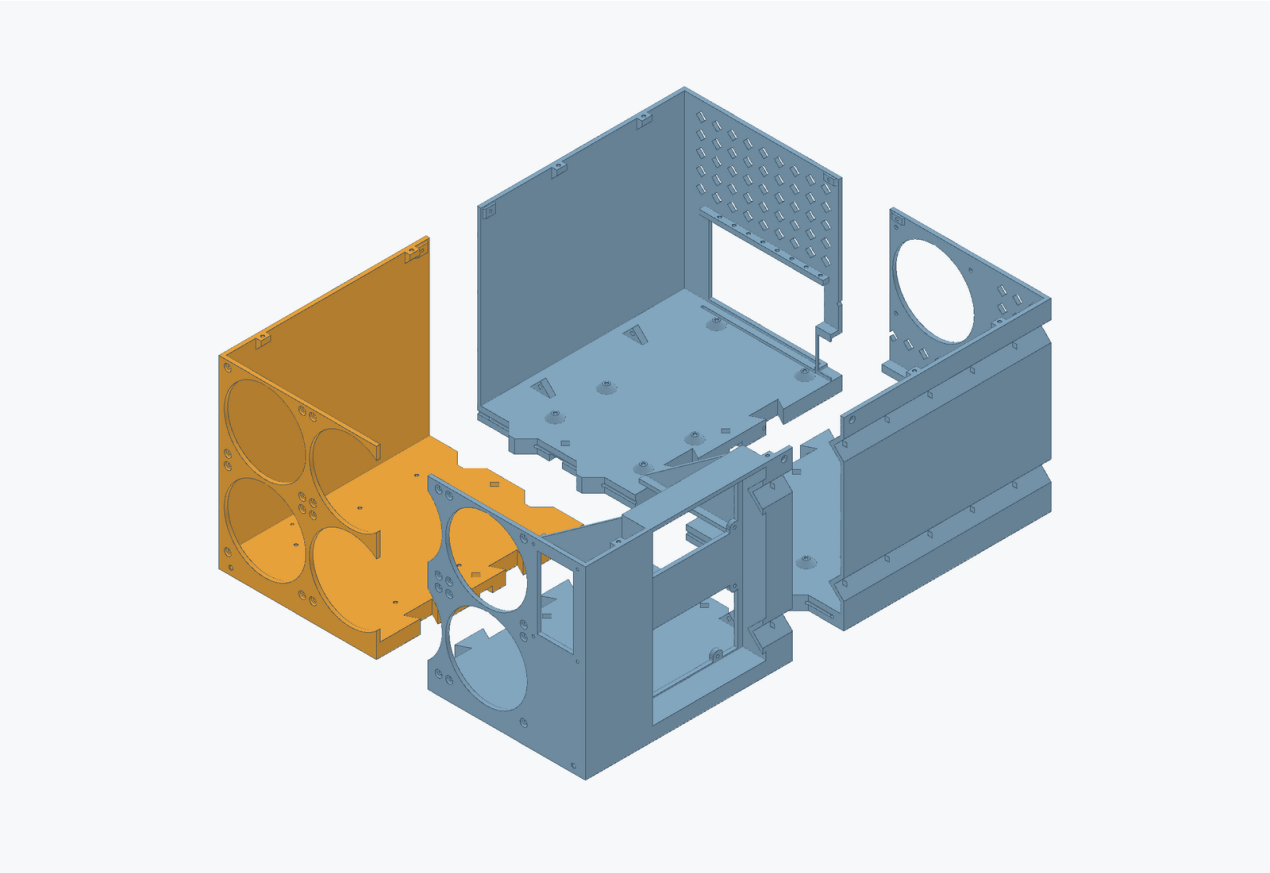
2. Seat the inserts flush. Support each boss or floor area, apply heat evenly and keep the insert square to its hole. Stop at the surface and let it cool before checking the M3 thread. Keep molten plastic out of the bore.

3. Check the structure separately. The bridges, wall straps and rod stops use through-screws, nuts and washers. They do not use the heatsets listed above. Keep their recesses free of debris.

JOIN THE SHELL

Lower the quadrants together

The dovetails and tapered tongues engage vertically in the assembled chassis orientation.



Exploded CAD: four quadrants spread apart for identification. Gold is the front-left section with its six floor pockets. The illustration shows separation, not a horizontal insertion path.

1. Set down rear-right. Support the rear-right base flat. Keep the rear I/O edge, cable-channel walls and upper rim clear of the bench supports.

2. Join rear-left. Align the centre dovetail profiles and lower rear-left vertically into rear-right. Support both bases so the joint does not carry a cantilevered quadrant.

3. Add front-right. Align the front/rear locating tongues with the rear-right section, then lower it into place. Keep the PSU shelf and front wall clear while handling it.

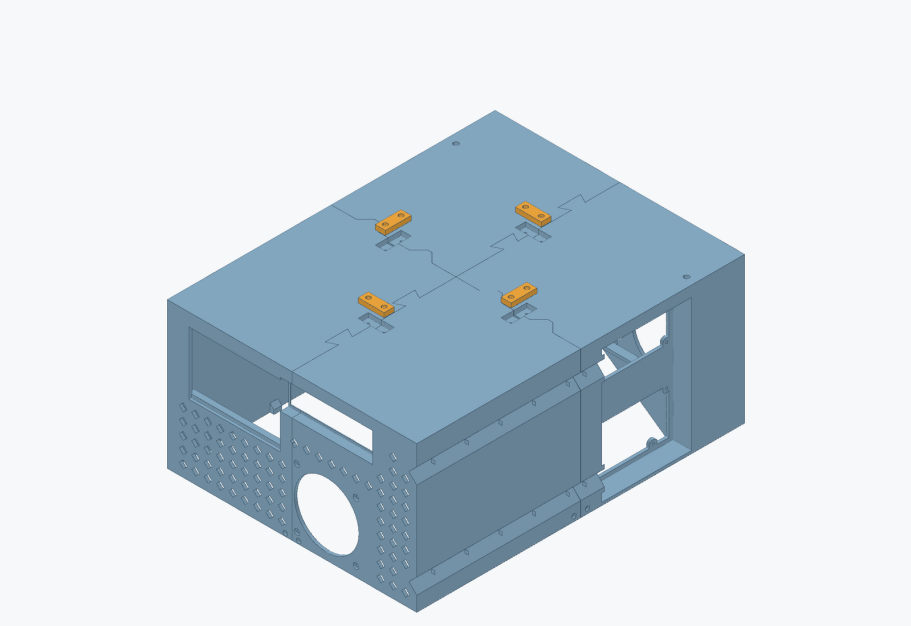
4. Add front-left last. Align both its rear and right-hand joints before lowering. Work all mating profiles together rather than forcing one fully home while another is misaligned.

> **Check before locking the seams**
> The four floors should sit in one plane. The modeled straight seam gap is 0.30 mm; do not use screws to crush that gap closed. If a joint binds, lift it back out and find the local interference before proceeding. Fit the connecting hardware before loading the shell.

LOCK THE FLOOR

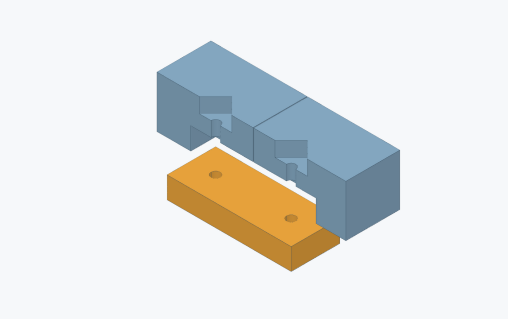
Fit four recessed base bridges

Use M3 x 14 screws here. The older M3 x 20 selection is superseded.



Underside CAD: gold bridge plates are pulled away from their four matching recesses. The complete empty chassis is turned over for access.

SECTION / BRIDGE JOINT



Fastener direction
From below: screw head - washer - printed bridge and chassis - washer - nut in the interior diamond pocket.

Two screws per bridge.
Eight M3 x 14 screws, eight nuts and sixteen washers total.

1. Support the empty shell upside down. Use padded blocks under robust wall/rim areas, avoiding the thin I/O edge, fan screw lands and PSU shelf. Fit each bridge into the recess that matches its orientation.

2. Start both fasteners. Place a washer under each screw head and insert from the underside. Put the other washer and nut into the recessed diamond pocket on the interior floor. Start both nuts before snugging the plate evenly.

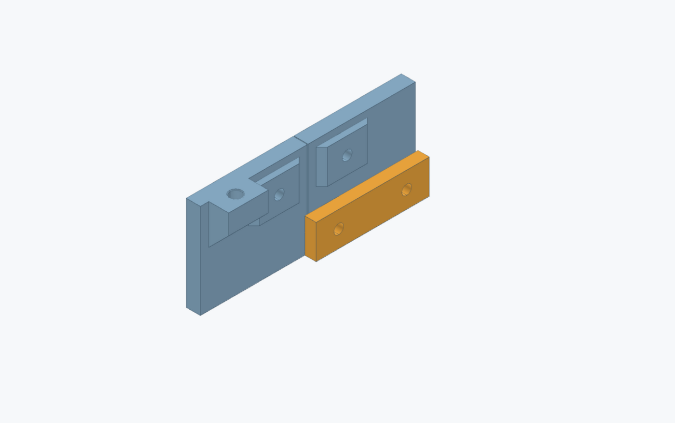
3. Inspect from inside. Every nut, washer and screw tip must remain below the floor surface. With the specified hardware, the model keeps nut tops 2.60 mm below the floor and the highest possible bolt tip at least 1.80 mm below it.

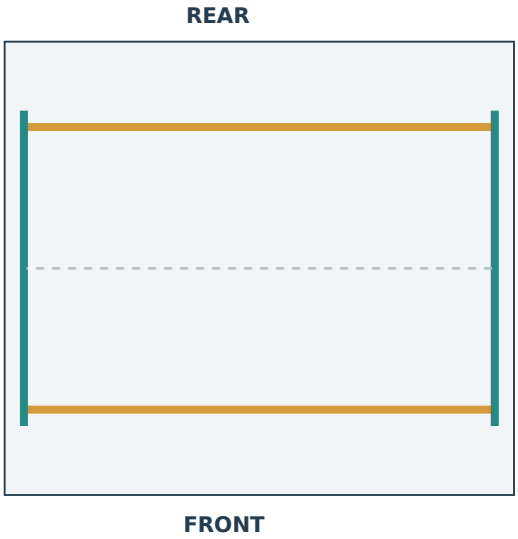
Do not substitute longer bridge screws under the motherboard. The washer/nut recess is intentional: nothing should protrude into the board underside clearance.

FINISH THE REINFORCEMENT

Three straps. Four rods.

Your middle crosswise rod channel stays empty. All remaining rods share the same height in the base.





Left: actual upper side-strap detail, pulled inward for visibility. Right: rod channel map; gold = front/rear crosswise, teal = longitudinal, dashed = omitted centre. Rod end positions depend on your longer stock.

1. Attach all three upper straps. Fit the left and right straps across the sidewall seams and the rear strap across the rear centre seam. Use two M3 x 12 screws per strap, with heads recessed outside and nuts/washers inside. Six screws, six nuts and twelve washers total.

2. Insert the two crosswise rods first. Feed the front and rear rods from the LEFT side toward their blind stops. Leave the middle channel vacant. Use a blunt pusher and hand pressure; do not hammer the printed joints.

3. Insert both longitudinal rods. Feed from the FRONT. Check that they pass the crosswise tips and seat without forcing the base apart. The design uses 4 mm rods in 4.3 mm channels; your longer rod ends must be checked physically.

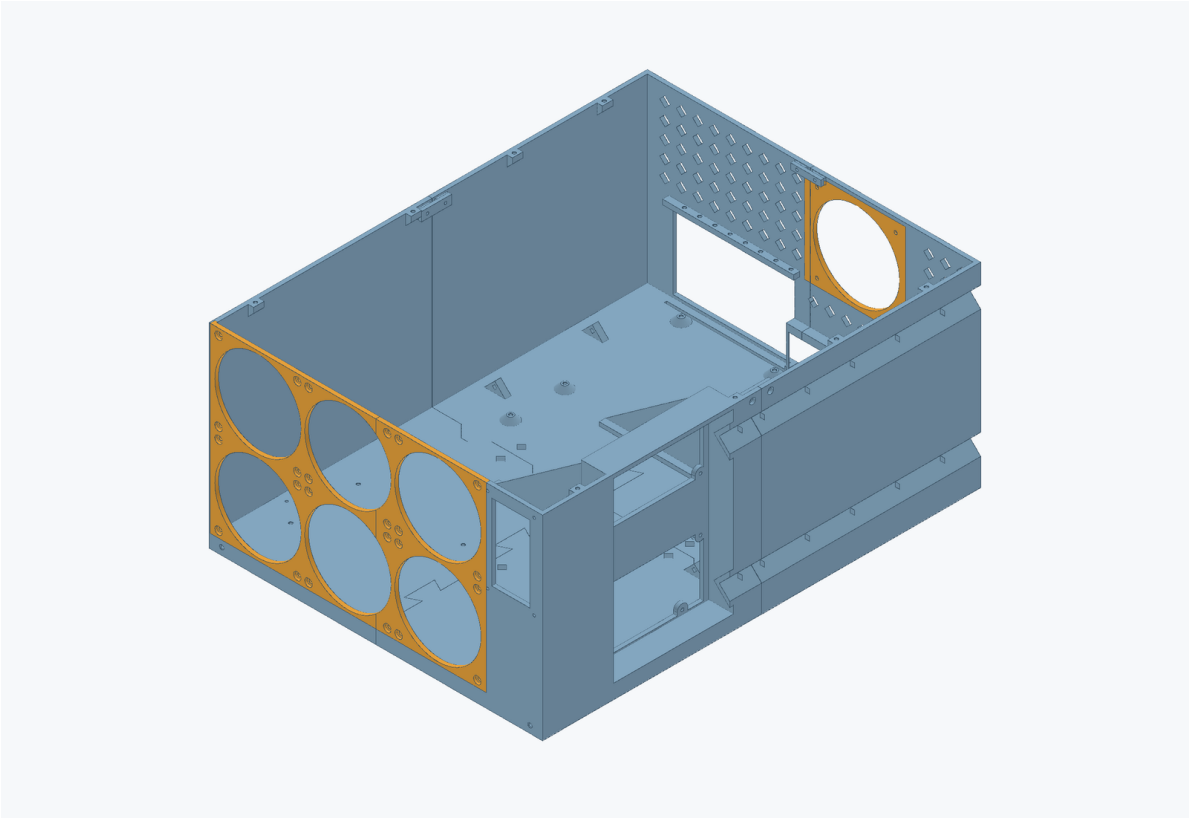
4. Fit the two rod-stop bolts if clear. The stops sit immediately ahead of the longitudinal rod ends. Insert M3 x 18 screws with washers downward from the interior; fit nuts and lower washers in the underside wells. If a longer rod blocks a stop, do not force the bolt through it. Resolve that retention conflict before handling a loaded chassis.

Your four-rod arrangement
The modified rod layout is recorded, not strength-certified. Keep the full base supported on the bench and rack shelf. Verify rod containment, closed seams and all seven connectors; avoid lifting a loaded chassis by its walls.

COOLING MOUNTS

Install six intakes and one exhaust

All seven chassis fans mount on the inside. Their short mounting screws enter from outside.



Actual CAD: gold marks the front fan wall and rear CPU-fan region. Exterior screw recesses leave 2 mm thick bearing lands at the mounting holes.

1. Front intake bank. Fit six 120 mm fans in the existing packed 3-wide x 2-high array at the left of the front panel. Airflow points into the chassis. Keep the right-hand panel/PSU area clear.

2. Use the recessed screw seats. Insert the appropriate fan screws from outside, through the recessed 4.5 mm clearance holes, into each fan frame. There are four screws per fan: 24 front and 4 rear. Confirm actual screw engagement; stop when the fan is secure without distorting the 2 mm plastic lands.

3. Bridge the front seam. The middle front fan column spans the quadrant seam. Install both of those fans before treating the front wall seam as bridged; there is no separate upper front strap.

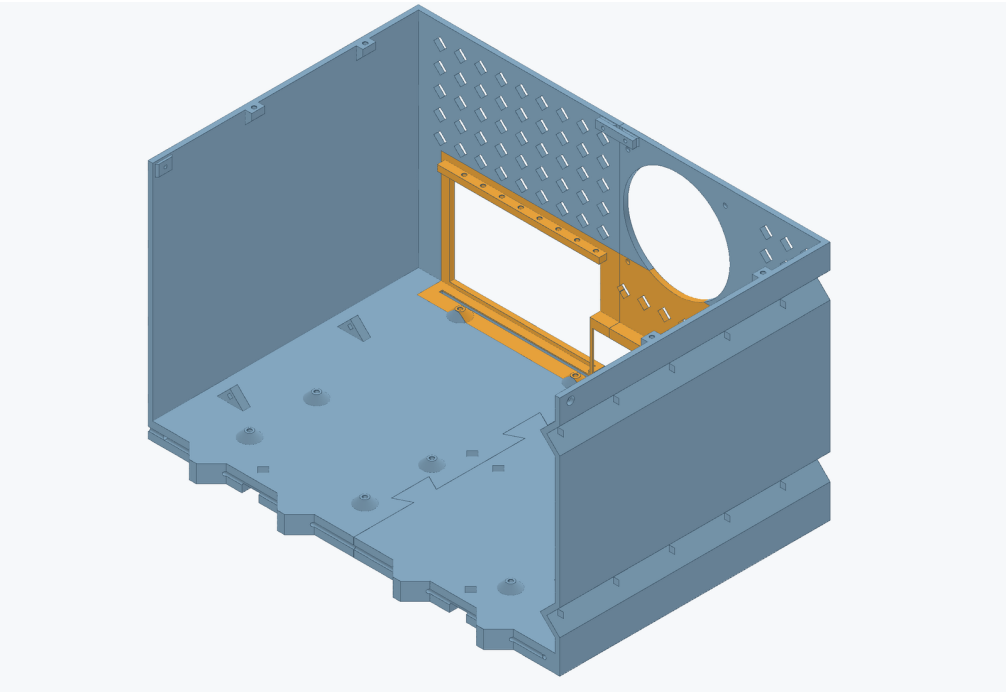
4. Rear CPU exhaust. Fit the 120 mm fan above the motherboard I/O opening with airflow out of the rear. Keep its cable out of the rotor and away from sharp printed edges.

Fit check: the 240 mm front stack leaves only 1.3 mm nominal clearance below the rim. Turn every rotor by hand and check the actual frame, wire exits and top clearance. The chassis fan mounts use the fans' suitable coarse-thread screws; the other structural fasteners are M3.

REAR INTERFACE

Fit I/O, motherboard and cards

Finish the recessed base fasteners before installing the motherboard.



Gold highlights the actual rear I/O and PCIe mounting region. The continuous lower card-tab channel is part of the existing floor; the large opening spans all eight positions.

1. Fit the I/O shield first. Seat the actual shield in the thin 1.2 mm mounting edge. Check its clip fingers and connector openings from both sides. The shield opening is approximately 158.75 x 44.45 mm.

2. Lower the motherboard onto the nine standoffs. Align all nine holes without dragging the PCB over screws or brass inserts. The custom standoffs use M3 heatsets; M3 x 5 screws are the documented starting length. Check actual engagement and keep the board flat as the screws are snugged.

3. Seat each GPU independently. Guide its lower rear bracket tabs into the continuous channel while aligning the PCIe connector. Check level, full connector seating and the latch before aligning the top bracket retention screw. Eight M3 insert positions are available on the ledge; use those corresponding to the actual brackets.

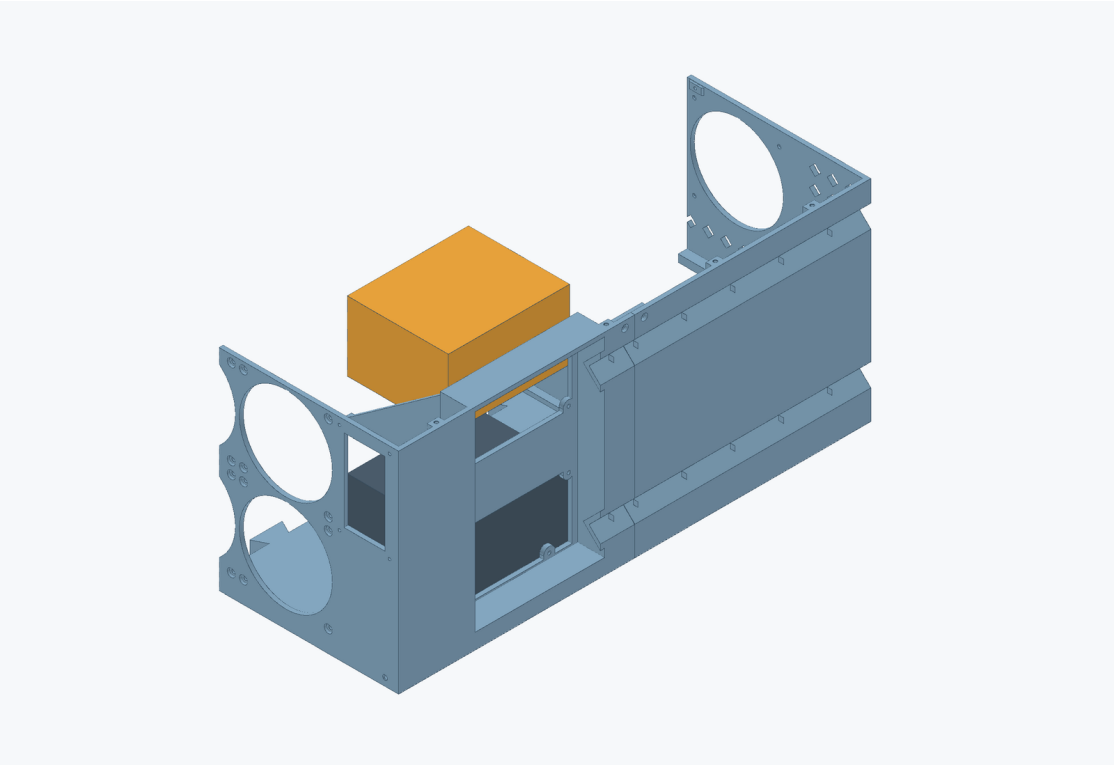
> **Known top-flange clearance issue**
> You reported insufficient rearward clearance at the bracket top flange. Check this before tightening any retention screw. Do not use a screw to pull the bracket backwards or the card down into its slot. If the flange prevents full seating or secure retention, stop that fitment and identify the contact. This guide makes no modification to the printed chassis or GPU bracket.

The rear retention screw starting selection is M3 x 5; engagement depends on the actual bracket and any washer. Do not borrow #6-32 PSU screws for these M3 inserts. Keep the eight-slot opening and diamond vents clear of wiring.

POWER-SUPPLY BAY

Install the lower and upper PSUs

Both exhaust through the recessed right-side panel; their intake fans face one another.



Actual right-side chassis CAD with simplified PSU envelopes. The gold upper PSU is lifted only to reveal the open-centre shelf; it rests on the two rails in service.

1. Lower PSU: intake up. Place the lower supply on the chassis floor with its exhaust/IEC face toward the recessed right wall. Its large intake fan faces upward into the gap between the supplies. Align the exhaust-face holes without twisting the chassis.

2. Upper PSU: intake down. Lower the upper supply onto the two shelf rails. Its intake fan faces downward into the same central gap. Check that both case edges bear on the rails and the fan grille clears the open centre and gussets.

3. Fasten the exhaust faces. Use four #6-32 UNC PSU screws per supply, eight total. Use the PSU maker's supplied or specified screws and length. These holes use a different thread from the chassis M3 heatsets. Support each PSU while starting the screws; do not use them to pull a badly aligned case onto the shelf.

4. Route the C13 power cords. The exhaust faces are recessed about 38.1 mm (1.5 in) from the side profile. Lay one cord into each open V-channel along the right side toward the rear. Thread ties through the diamond passages, behind and around the cable; the connector itself need not pass through a channel.

Keep the two PSU intakes unobstructed and the cords free of sharp bends or pinching at the recess. Confirm the actual C13 connector and bend stay within the desired rack profile. There are six tie stations per channel; 2.5 mm ties were the starting allowance. Snug them enough to retain the cord without crushing its jacket.

FINISH THE SHELL

Cable routing and final inspection

This guide ends at the chassis. The separate duct guide covers the duct supports, fan bank and GPU interfaces.

Built-in provisions

Feature	How to use it
Six internal cable-tie lugs	Route low-voltage cables around the perimeter. Leave service slack without letting loops reach a rotor or card connector.
Rearward PSU cord channels	Use the two open V-grooves and their diamond tie passages. Keep the rear cord exits accessible.
Top-front-right button opening	A future external panel fastens from the inside through four M3 clearance holes. The opening is about 2 x 4 in; panel, switch and screw lengths are not supplied in the chassis set.
Eight top-rim inserts	Reserved for a future lid. Choose screw length when lid thickness is known; do not let a screw bottom in a blind insert.
Six front-left floor inserts	Reserved mounting points. Keep unused inserts flush and free of debris.

Before the chassis is loaded or powered

[ ] Base and seams. All four quadrants sit flat; four underside bridges and three upper straps are secure. Nuts, washers and bolt tips remain recessed under the motherboard.

[ ] Rods. Only the front/rear crosswise rods and two longitudinal rods are installed. All ends are contained and the actual stop-bolt retention is confirmed with your longer stock.

[ ] Fans. Front six blow inward; rear CPU fan blows outward. All rotors turn freely, cables are clear, and the thin screw lands are not cracked or distorted.

[ ] Motherboard and GPUs. Board rests on all nine standoffs. Cards are fully seated and retained without force against the known top-flange obstruction.

[ ] PSUs. Lower intake up, upper intake down; both exhaust right. Upper supply bears on both rails. Eight correct PSU screws are fitted and the central intake gap is open.

[ ] Cables and support. No loose fasteners, swarf or trapped leads remain. Use only the modular cables supplied/approved for each PSU. The rack shelf supports the base; do not suspend the printed chassis by its sidewalls.

Source: current Rev24 dimensions and Step16 chassis CAD, plus your October build notes. Figures are CAD views or labeled schematics; exploded offsets are illustrative. Four-rod strength, printed-part creep and the reported bracket interference require physical judgment during assembly.